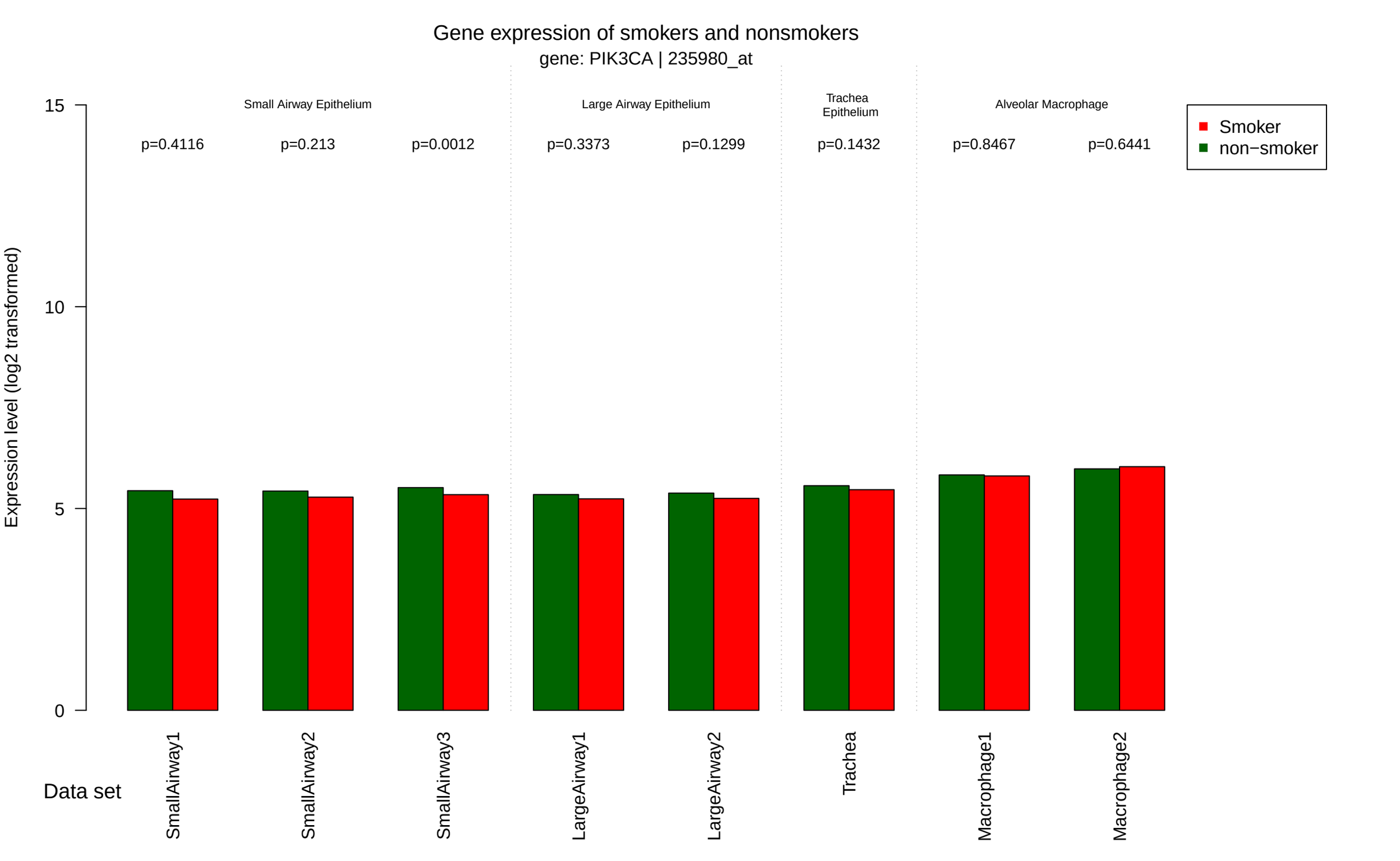
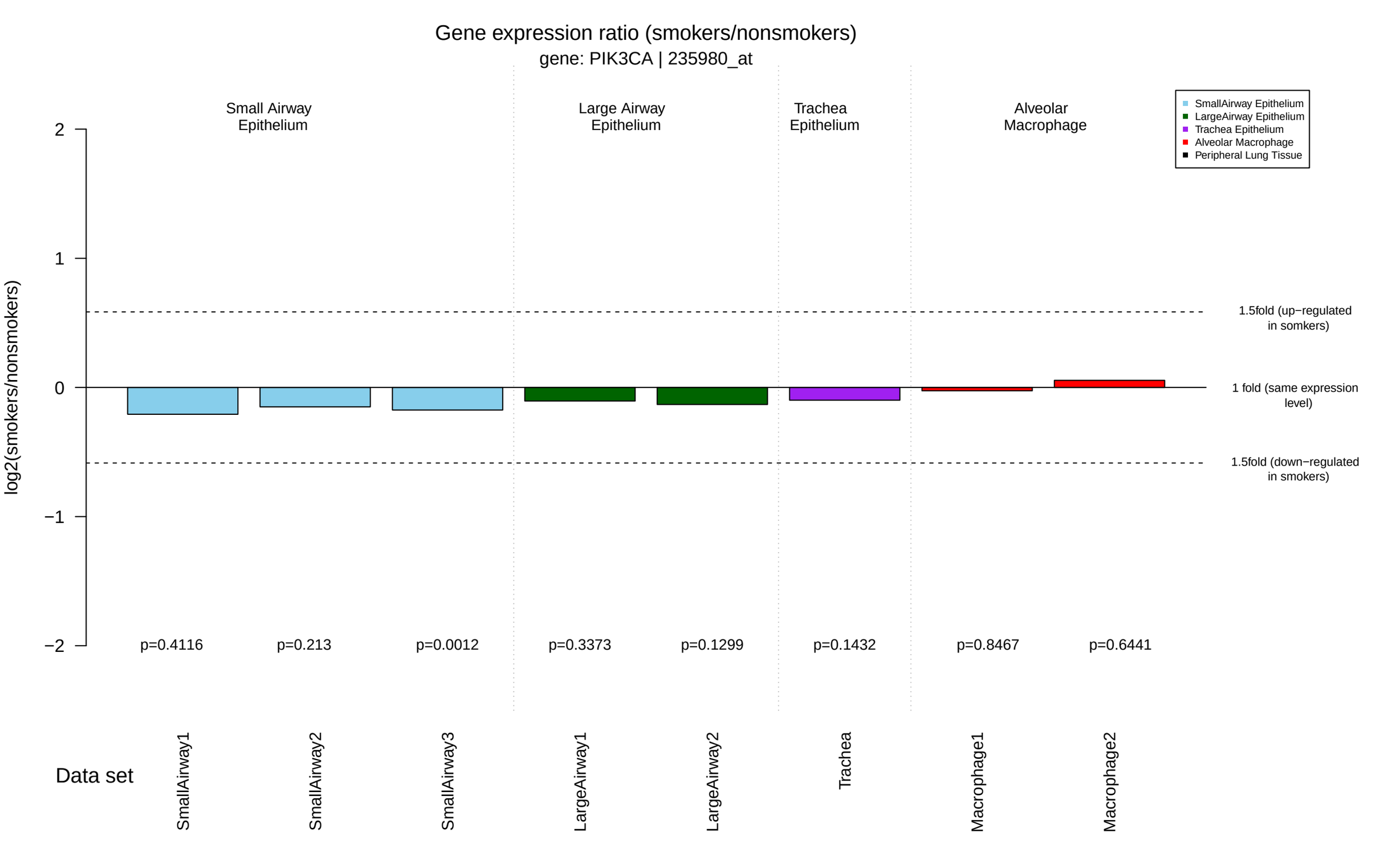
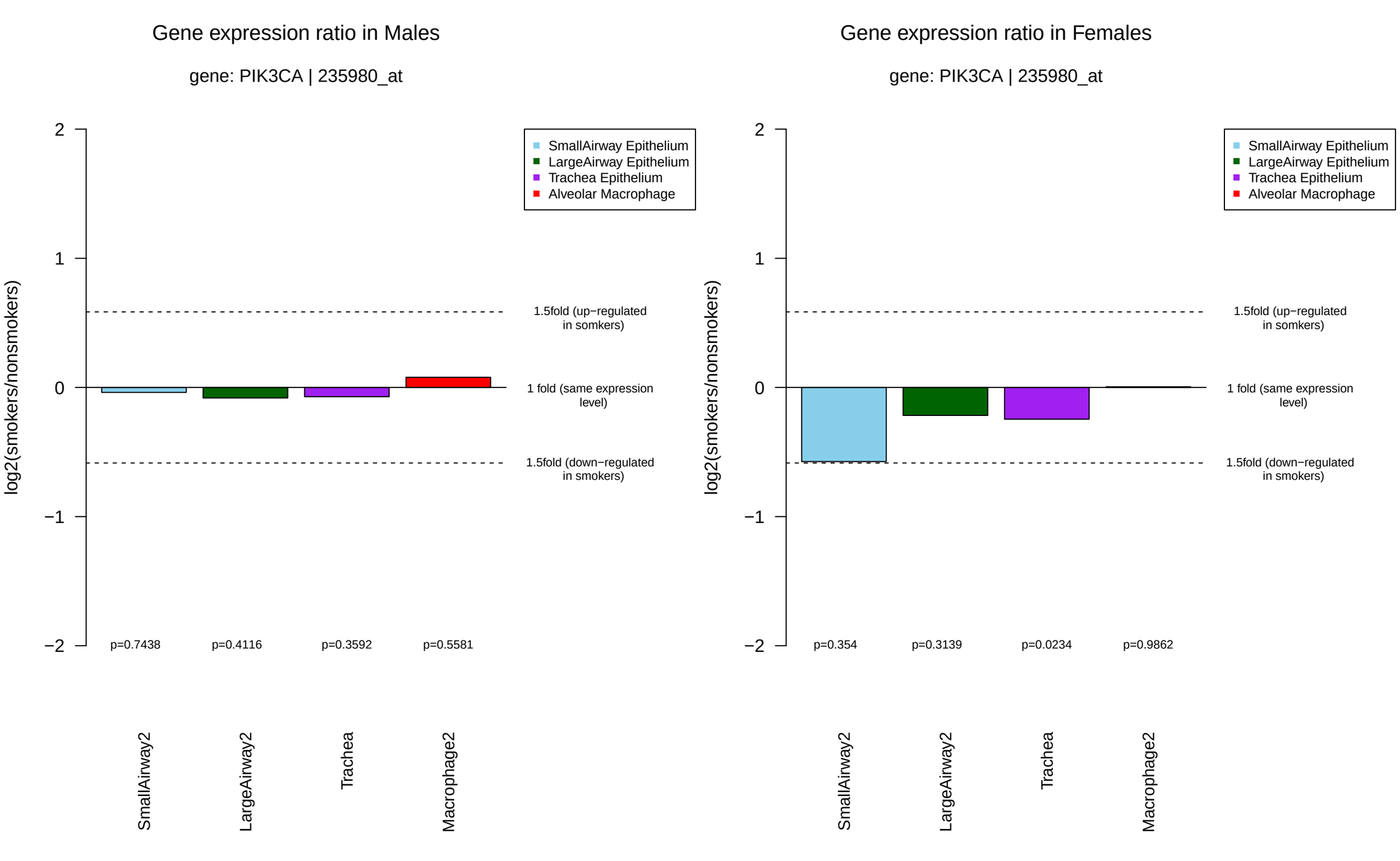
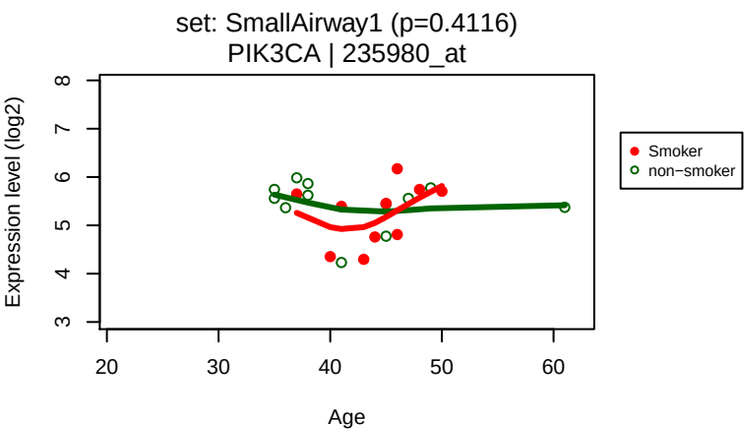
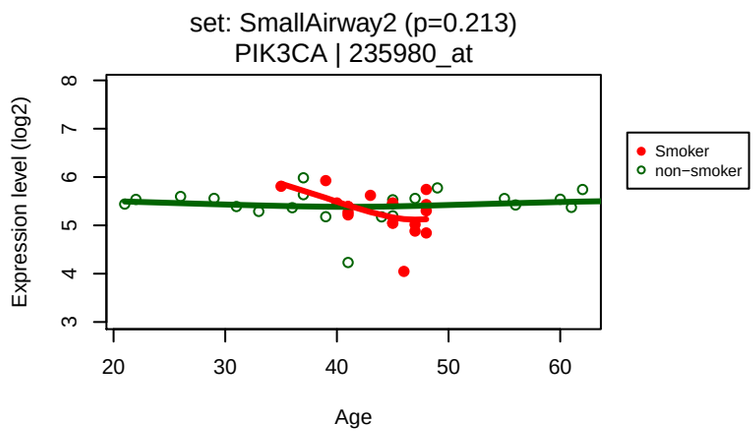
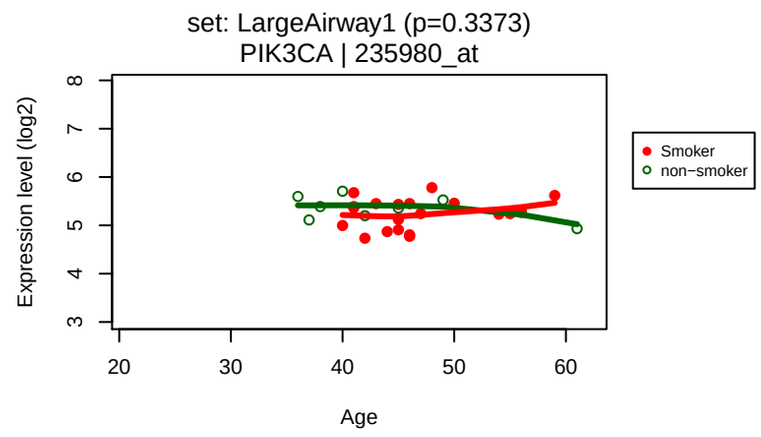

Gene expression of smokers and nonsmokers
gene: PIK3CA | 235980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4116
p=0.213
p=0.0012
p=0.3373
p=0.1299
p=0.1432
p=0.8467
p=0.6441
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

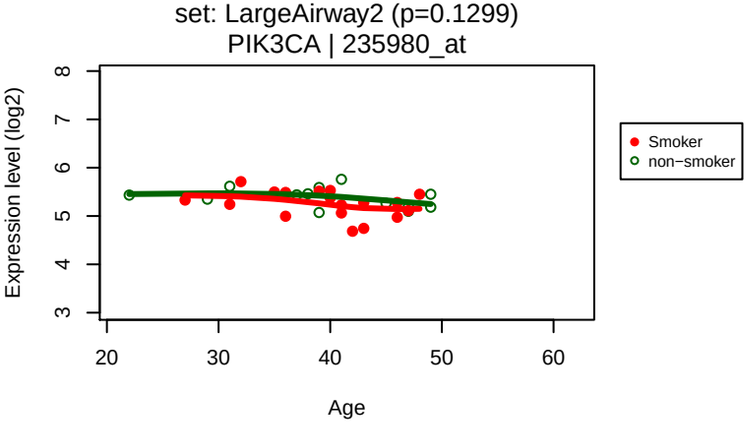

Gene expression ratio (smokers/nonsmokers)
gene: PIK3CA | 235980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4116
p=0.213
p=0.0012
p=0.3373
p=0.1299
p=0.1432
p=0.8467
p=0.6441
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

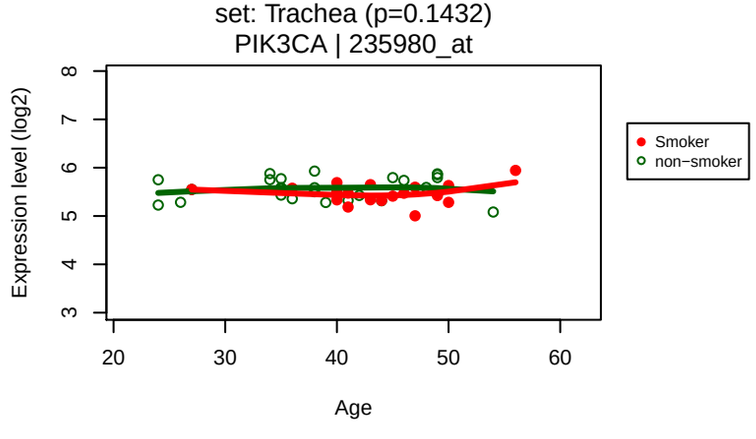

Gene expression ratio in Males
gene: PIK3CA | 235980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7438
p=0.4116
p=0.3592
p=0.5581
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3CA | 235980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.354
p=0.3139
p=0.0234
p=0.9862
SmallAirway2
LargeAirway2
Trachea
Macrophage2

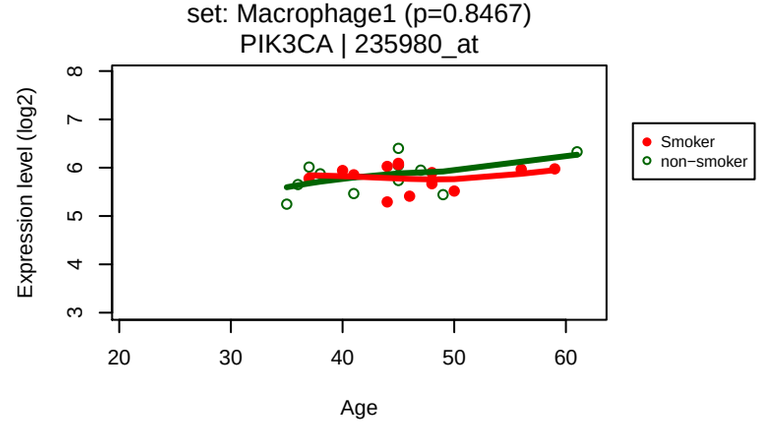

set: SmallAirway1 (p=0.4116)
PIK3CA | 235980_at
Expression level (log2)
Age
Smoker
non−smoker

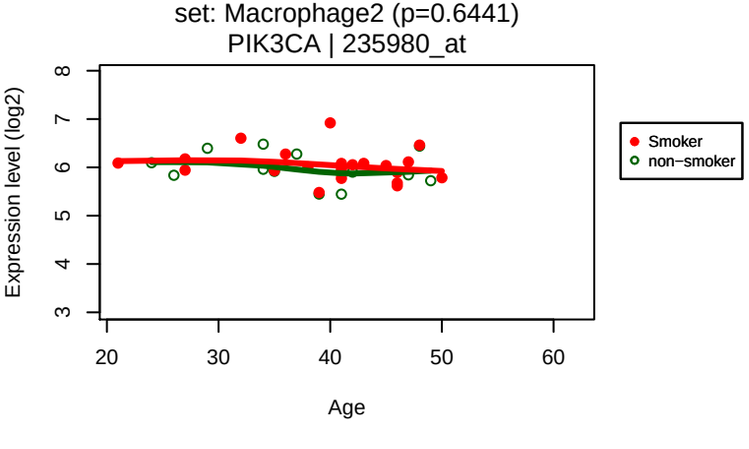

set: SmallAirway2 (p=0.213)
PIK3CA | 235980_at
Expression level (log2)
Age
Smoker
non−smoker

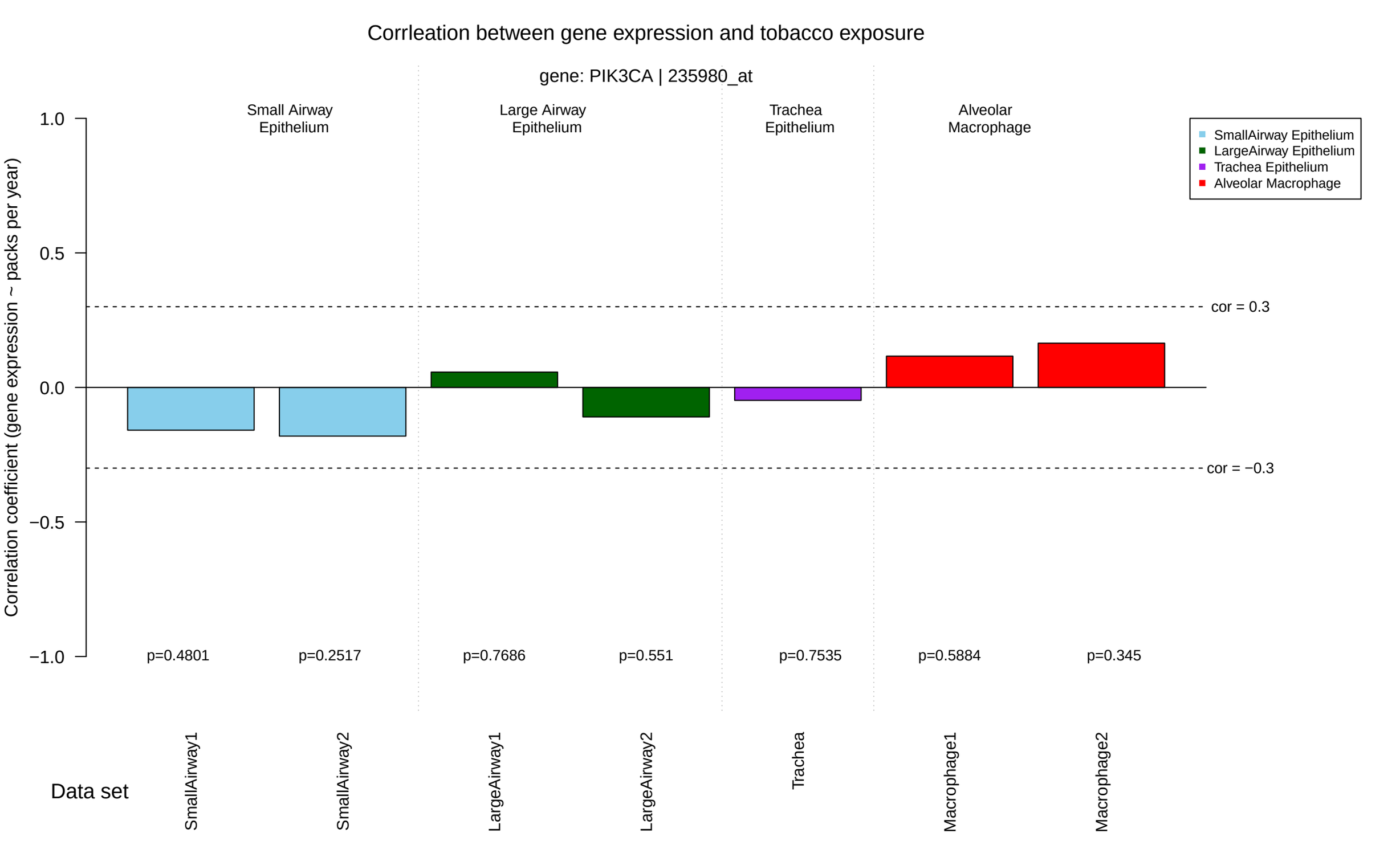

set: LargeAirway1 (p=0.3373)
PIK3CA | 235980_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1299)
PIK3CA | 235980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1432)
PIK3CA | 235980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8467)
PIK3CA | 235980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6441)
PIK3CA | 235980_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3CA | 235980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4801
p=0.2517
p=0.7686
p=0.551
p=0.7535
p=0.5884
p=0.345
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2